knows about it. But why should anyone else care? After all, quantum physics certainly works. For that matter, why should *physicists* care? Their mathematics makes accurate predictions; isn't that enough?

But science is about more than mathematics and predictions—it's about building a picture of the way nature works. And that picture, that story about the world, informs both the day-to-day practice of science and the future development of scientific theories, not to mention the wider world of human activity outside of science. For any given set of equations, there's an infinite number of stories we could tell about what those equations mean. Picking a good story, and then searching for holes in that story, is how science progresses. The stories told by the best scientific theories determine the experiments that scientists choose to perform and influence the way that the outcomes of those experiments are interpreted. As Einstein pointed out, "The theory decides what we can observe."

The history of science bears this out over and over again. Galileo didn't invent the telescope—but he was the first to think of pointing a good one at Jupiter, because he believed that Jupiter was a planet, like Earth, that went around the Sun. After that, telescopes were used regularly to look at everything from comets to nebulae to star clusters. But nobody bothered to use a telescope to find out whether the Sun's gravity bent starlight during a solar eclipse—not until Einstein's theory of general relativity predicted just such an effect, over three centuries after Galileo's discovery. The practice of science itself depends on the total content of our best scientific theories—not just the math but the story of the world that goes along with the math. That story is a crucial part of the science, and of going beyond the existing science to find the next theory.

That story also matters beyond the confines of science. The stories that science tells about the world filter out into the wider culture, changing the way that we look at the world around us and our place in it. The discovery that the Earth was not at the center of the universe, Darwin's theory of evolution, the Big Bang and an expanding universe nearly 14 billion years old, containing hundreds of billions of galaxies, each containing hundreds of billions of stars—these ideas have radically altered humanity's conception of itself.

Quantum physics works, but ignoring what it tells us about reality means papering over a hole in our understanding of the world—and